

STANDARD OPERATING PROCEDURE

for

Field Operation of

Digital Level

Module Name	
Module Code	
Maintained by	Faculty of Topo. & GIS
Version	1.0

Document Name	Standard operating procedure for Field Operation of Digital Level		
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1 Objective

Using two known Bench Marks (BM's), establish or densify additional Bench Marks within the Area of Interest (AOI) by running a leveling line. The barcode on the leveling staff is stored in the instrument as a reference signal. During measurement, the visible portion of the staff within the field of view is captured by the line decoder and converted into a measuring signal. This signal is compared with the reference signal to determine the height reading and horizontal distance. The staff must be held vertically (perpendicular to the ground), similar to conventional optical leveling procedures. With the provision of artificial illumination, accurate measurements can also be performed in low-light or nighttime conditions. The sensor operates effectively across a wide frequency range, from visible light to infrared.

2 Hardware & Software

2.1 Hardware:

- i. 01 no. Digital Level
- ii. 01 No. Tripod Stand
- iii. 01 No. Brass Brinjee
- iv. 02 No. Bar coded Staves
- v. 02 No. Iron brad (panja)

2.2 Software:

- i. Leica Infinity/Leica Geo office

3 Components of Leica LS10 Digital Level

Container Contents



010043.001

- a) Instrument
- b) Quick Guide/USB documentation card
- c) GKL311 battery charger (optional)
- d) GEB331 batteries (optional)
- e) Spare stylus (optional)
- f) Allen keys (1.5 mm/2 mm)
- g) GEV223 USB data transfer cable (optional)
- h) Car adapter cable for GKL311 (optional)
- i) GEV192 AC power adapter for GKL311 (optional)
- j) Sunshade
- k) Rain Cover

2.3

Instrument Components

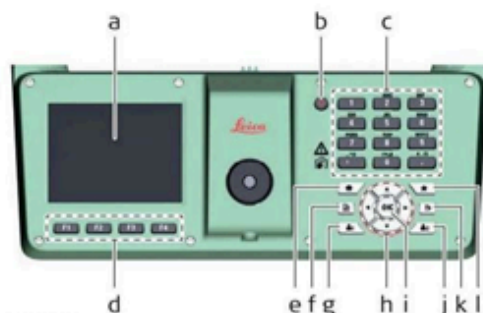
Instrument Components (Part 1)



010044.002

- a) Optical sight
- b) Handle with integrated prism for viewing the circular level
- c) Circular level
- d) Touchscreen
- e) Battery compartment, also containing interface for USB stick and Mini USB
- f) Horizontal drive
- g) Push button to open battery compartment
- h) Function keys
- i) ON/OFF key
- j) Keyboard
- k) Eyepiece
- l) Protective cap for crosshair adjustment screw
- m) Horizontal circle

Keyboard



010046.002

- a) Touchscreen
- b) ON/OFF key
- c) Alphanumeric keypad
- d) Function keys **F1** to **F4**
- e) Home key
- f) Page key
- g) User key 1
- h) Navigation keys
- i) Enter key
- j) User key 2
- k) ESC key
- l) Favourites key

Keys

Key	Description
	ON/OFF key to switch the instrument on or off or to set it into standby mode.
	Alphanumeric keypad for entering text and numerical values.
	Home key. Returns to the Main Menu .
	Page key. Displays the next screen when several screens are available.
	Favourites key. Quick-access to measurement supporting functions.
	ESC key (in general): Quits a screen or edit mode without saving changes. Returns to next higher level. ESC key (in all levelling applications): After confirming the action, the ESC key deletes the last observation and allows to repeat it.
	User key 1. Programmable with a function from the Favourites menu.
	User key 2. Programmable with a function from the Favourites menu.
	Navigation keys. Control the focus bar within the screen and the entry bar within a field.
	ENTER key. Confirms an entry and continues to the next field.
	Trigger key. Triggers a measurement. Programmable with functions which allow the auto-focussing of the telescope (only LS15), the height and distance reading and the storing of the measurement data. For details on how to define the functions of trigger key, refer to "6.1 Work Settings".
	Function keys that are assigned to the variable functions displayed at the bottom of the screen.

4 Pre-requisites

The following are required before commencing leveling:

- a. The surveyor should have theoretical knowledge of leveling
- b. Two known Benchmarks (BM) with proper descriptions are required to start the leveling line.
- c. A check leveling connecting these BMs must be carried out before starting fieldwork to verify the stability of the given BMs.

5 Precautions

5.1 Precautions For Levelling

i) The collimation error of the level should be checked and accepted only if it does not exceed **0.002 m at 100 m**.

ii) The instrument should be kept under shade for about **2 minutes before commencing work each day** to allow for automatic temperature adjustment, and it should remain shaded throughout the day.

iii) The instrument should always be placed midway between the two staves to minimize:

- Collimation error,
- Error due to the curvature of the Earth, and
- Error due to irregularity of the bubble tube curvature.

iv) The staves should be placed on **iron brads firmly fixed in the ground** to avoid gradual sinking, except when placed on Bench Marks (BM).

v) An **even number of stations should be observed between BMs**, ensuring that the same staff is used on all BMs.

vi) The instrument should always be set up such that a **particular leg faces a particular staff**, in order to minimize errors caused by the natural tendency of the instrument to settle in a given direction.

vii) **Readings below 0.300 m and above 2.700 m** on the staff should be avoided to

minimize the effects of atmospheric refraction and non-verticality of the staff, respectively.

viii) Parallax must be **completely eliminated** before commencing observations.

ix) Daily work should always be **closed on a BM or temporary BM (TBM)**.

x) The accuracy limit for **Single Tertiary Levelling** is: $\pm 0.025\sqrt{K}$ Metres, where K is the distance in kilometres.

xi) The accuracy limit for **Double Tertiary Levelling** is: $\pm 0.0125\sqrt{K}$ Metres, where K is the distance in kilometres.

5.2 Precautions For Digital Levelling

In addition to the general precautions for levelling, the following specific precautions apply to digital levels:

- a. The **Invar staff must be cleaned regularly**. Extra care should be taken during use and transportation, as it is a very sensitive instrument.
- b. The **Invar staff should be kept either fully in light or fully in shade**, as sudden changes in brightness may affect the readability.
- c. The **line of sight must always be clear**, with no obstructions between the digital level and the staff.
- d. The **battery must be fully charged after completion of work each day** to ensure uninterrupted operation.
- e. The accuracy limit for **High Precision Levelling** is: $\pm 0.003\sqrt{K}$ Metres, where K is the distance in kilometres.
- f. The **focusing on the staff must be sharp and clear** before taking any observation.

6 Roles & Responsibilities

The surveyor will have the following roles and responsibility.

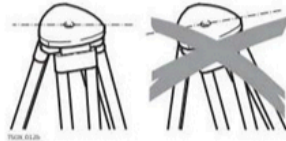
- The surveyor will identify the BMs on the ground based on the records and documents available with him.
- Once the BMs have been identified on the ground, carry out Levelling line to establish the BMs on AOI

7 Setting up the LS10 Digital Level

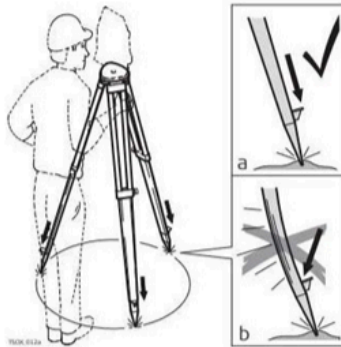


For highest accuracy levelling tasks, use a tripod with fixed legs, for example 328422 GST40.

Tripod

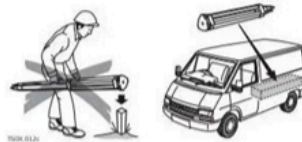


When setting up the tripod pay attention to ensuring a horizontal position of the tripod plate. Slight corrections of inclination can be made with the foot screws of the tribrach. Larger corrections must be done with the tripod legs.



Loosen the clamping screws on the tripod legs, pull out to the required length and tighten the clamps.

- In order to guarantee a firm foothold sufficiently press the tripod legs into the ground.
- When pressing the legs into the ground note that the force must be applied along the legs.



Careful handling of tripod.

- Check all screws and bolts for correct fit.
- During transport, always use the cover supplied.
- Use the tripod only for surveying tasks.

Setup Step-by-Step Setting up the Instrument on the Tripod

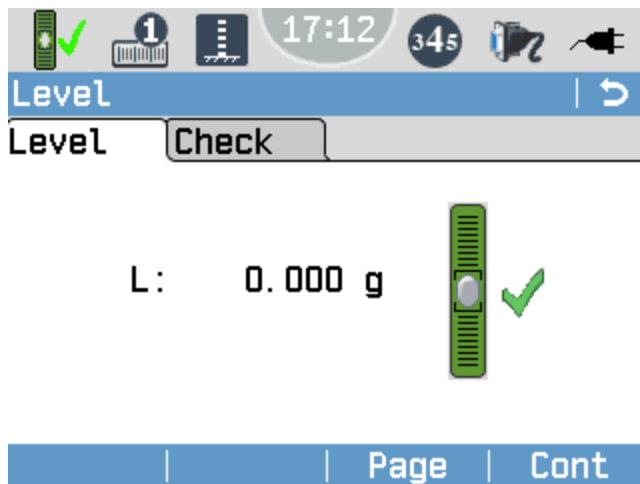
Step	Description	
1.	Set up the tripod.	
2.	Extend the tripod legs to allow for a comfortable working posture.	
3.	Fasten the instrument onto the tripod.	
4.	Tighten the central fixing screw.	

010756.001

7.1 Setting up of the Tripod

- Extend the tripod legs to a stable position.
- Adjust the tripod height so that the instrument is at a comfortable working level.
- Firmly place the tripod on the ground and ensure it is stable. It should be levelled with foot screws.

Switch on LS10 and Level the circular bubble then press Cont.(F4)



7.2 Mounting the Digital Level

- Carefully place the Levelling machine onto the tripod.
- Secure it using the central fixing screw.

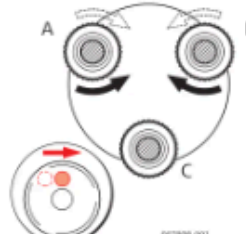
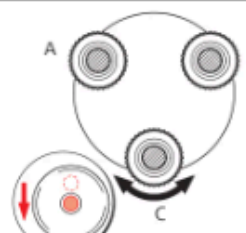
7.3 Setting up of levelling Staves

- Set up the Levelling staves in a vertical position.
- The perpendicularity of staff to be ensured by levelling the staff properly using the bubble.
- Ensure that there is no obstruction between the staff and levelling machine.

7.4 Levelling the Digital Level Instrument

Centring the Circular Level Manually

The circular level helps you to level up the instrument manually. To level up the instrument precisely, you can also use the electronic level. Refer to 4.4" For LS15: Level Up with the Electronic Level Step-by-Step".

Step	Description	
1.	Turn the instrument, so that the ocular is positioned above footscrew C.	
2.	Turn the footscrews A and B simultaneously in opposite directions until the bubble is on the middle axis of the circular level.	
3.	Turn the footscrew C until the bubble is centred.	

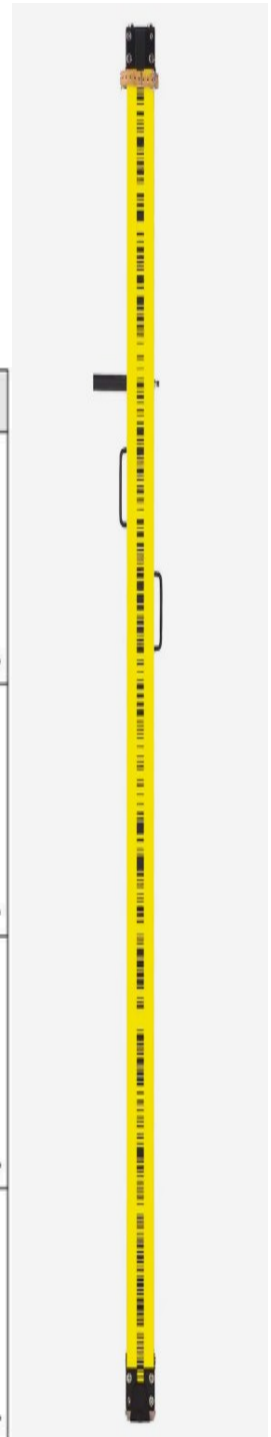
Adjusting the Reticle to the Eyesight of the User

Step	Description
	To ensure a parallax free sight for optical readings of a staff, you need to adjust the reticle to the eyesight of the user.
1.	Point the telescope towards a bright background.
2.	Turn the ocular until the reticle is focussed and appears sharp and black.

- Use the circular bubble level to approximately level the instrument.
- Adjust the tripod legs as necessary.
- Fine-tune the level using the three leveling screws until the bubble is centered in the circular vial.
- Turn the screws in opposite directions, in pairs, to maintain balance and avoid excessive tilting.
- Make small, gradual adjustments to prevent overshooting.
- Recheck the level after all adjustments to ensure accuracy before proceeding.

➤ Setting up of levelling Staves

Icon	Description
	The orientation of the staff is set to upright. It is only possible to take measurements with the 0-mark at the bottom of the staff. Tap the icon to change the orientation to inverse. The earth curvature correction is set to Off. Refer to "6.2 Regional Settings" for details on how to set the earth curvature correction to On.
	The orientation of the staff is set to inverse. It is only possible to take measurements with the 0-mark at the top of the staff. The measured values are negative. Tap the icon to change the orientation to upright. The earth curvature correction is set to Off. Refer to "6.2 Regional Settings" for details on how to set the earth curvature correction to On.
	The orientation of the staff is set to upright. It is only possible to take measurements with the 0-mark at the bottom of the staff. Tap the icon to change the orientation to inverse. The earth curvature correction is set to On. Refer to "6.2 Regional Settings" for details on how to set the earth curvature correction to Off.
	The orientation of the staff is set to inverse. It is only possible to take measurements with the 0-mark at the top of the staff. The measured values are negative. Tap the icon to change the orientation to upright. The earth curvature correction is set to On. Refer to "6.2 Regional Settings" for details on how to set the earth curvature correction to Off.



8 Basic Settings in a Digital Level

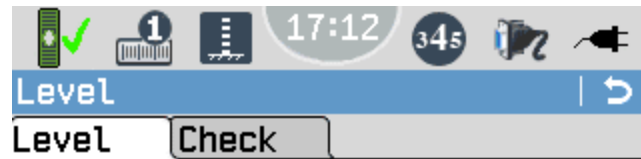
8.1 Importance of Basic Settings in a Digital Level

Basic settings in the Digital level are critical for ensuring accuracy, efficiency, and reliability during leveling operations. These settings act as the foundation on which all measurements are recorded and stored.

1. **Accuracy of Data** – Correct configuration of units, date, and time ensures that field observations are precise and properly referenced for further processing.
2. **Consistency in Projects** – Using standardized units and formats avoids confusion when data is shared among different users or imported into processing software.
3. **Traceability** – Accurate date, time, and job information provide a reliable audit trail, which is essential for quality control and future reference.
4. **Ease of Operation** – Proper display settings (brightness, contrast, sound) make the instrument user-friendly under varying field conditions, reducing chances of operator error.
5. **Efficient Power Use** – Battery and power-saving settings help extend working hours, minimizing interruptions during long fieldwork sessions.
6. **Smooth Data Transfer** – Communication settings ensure compatibility with computers and office software, enabling seamless transfer and reporting.
7. **Instrument Reliability** – Regularly checking instrument information (firmware, memory, serial number) helps in troubleshooting, maintenance, and ensuring compliance with standards.

In summary, **basic settings are not just initial configurations but essential controls that directly influence measurement quality, field productivity, and project reliability.**

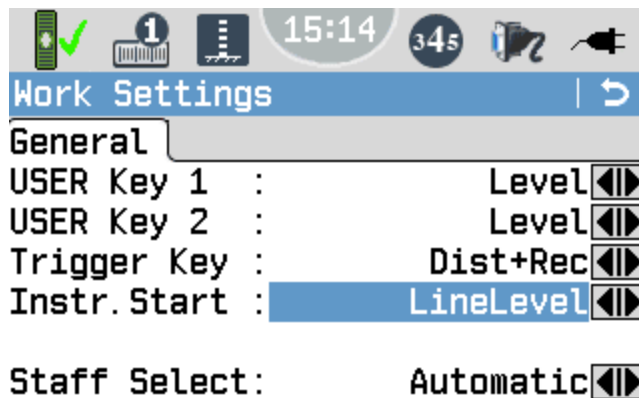
8.2 Basic Settings (need to be done only once you Switch on LS10 first time.)



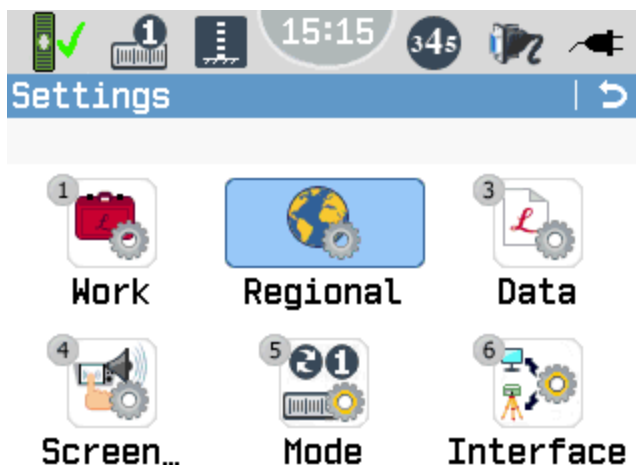
L: 0.000 g



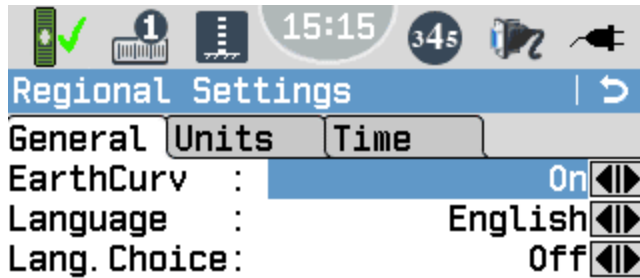
- Then Select Work and Make following settings



- Then Cont.
- Then Select **Regional**

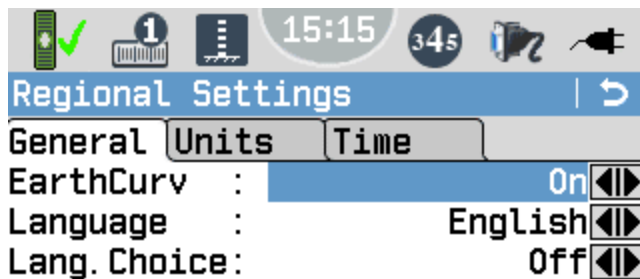


- And make following settings in General Tab



Default | Delete | Page | Cont

- Then Page Change (F3) for Unit Page
- And make following settings
- And Cont.

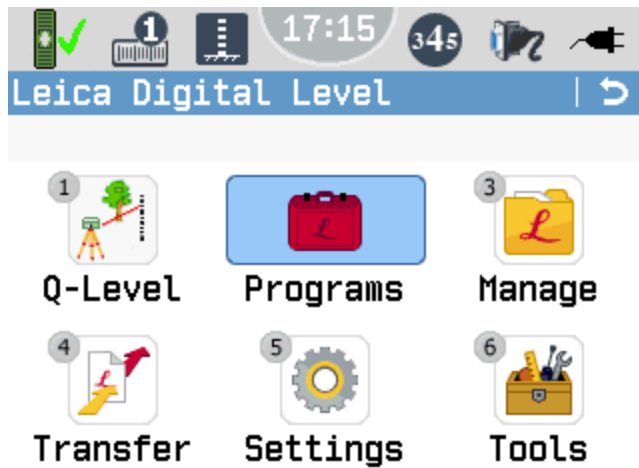


Default | Delete | Page | Cont

9 Creating a Job in Digital Level

9.1 Creation of a Job

- Highlight on **Programs** and Click **ok** button



- Then Select **Line Level** and Click **Ok**.



- Press **F1** to Create a New Job and Press **F1** again to enter the Desired Job Name.

17:18 34s

Select Job | ↩

Data

Job : 123

Operator: -----

Remark 1: -----

Remark 2: -----

Date : 12. 08. 2019

Time : 15:37:09

New | Cont

- Enter the Job Name and click **ok**, if it is required one can input a operator name & remark then **Cont**.

17:19 ABC

Enter Job Data | ↩

Data

Job : TEST

Operator: -----

Remark 1: -----

Remark 2: -----

Date : 10. 09. 2019

Time : 17:19:32

Back | Cont

9.2 Setting of Tolerance

- Then Click **F2** to **set tolerance**

17:21

Set Tolerance

Config.	Values
Precise	On
Min. Distance:	On
Max. Distance:	On
Dist B-F	On
TDist. Bal	On
Staff Ends	On
Stn. Diff	On

Page | Cont

Press F3 to set the desired value as per your project requirement and then **Cont.**

(Below attached Values are by default values.)

17:23

Set Tolerance

Config.	Values
Min. Distance:	4.000 m
Max. Distance:	50.000 m
Dist B-F	1.000 m
TDist. Bal	3.000 m
Staff Top	2.50000 m
Staff Bottom:	0.50000 m
Stn. Diff	0.00030 m

Reset | Page | Cont

- For DT/HP leveling work, following setting for value should be inserted

17:13

Set Tolerance

Config.	Values
Min. Distance:	2.000 m
Max. Distance:	60.000 m
Dist B-F	2.000 m
TDist. Bal	4.000 m
Staff Top	2.70000 m
Staff Bottom:	0.30000 m
Turn. Diff	0.00050 m

Reset | Page | Cont

Staff Bottom: 0.30000 m
 Stn. Diff : 0.00090 m
 Turn. Diff : 0.00050 m
 Reset | Page | Cont

- Then Press **F4** Start.
- Give the Line Name and Select method like **BFFB(Double Tertiary levelling)**
- Give the name of the bench mark point in (Pt ID).
And, input its known height if any, in (Start H).
Then Click Set.

Line Levelling | ↻
 Line
 Name : LINE00001
 Method : BFFB
 PtID : A1
 Start H: 0.00000 m
 Staff1 : P1
 Staff2 : P2
 Back | List | Set

Then in below screen **B** of BFFB is highlighted in **RED** color here one to sight ,Back sight point

Line Levelling | ↻
 Meas Code
 Backsight: **B**FFB BFFB
 Stat. ID : Stn1
 PtID : A1
 Rem. : -----
 Height : 0.00000 m
 Rec | Page | Back

10 Collimation Checking Procedure for Leica Digital Level (A-X-X-B Method)

10.1 Objective

- Collimation checking should be carried out before starting the levelling line to determine and correct the collimation error of a Leica digital level using the A-X-X-B method.

10.2 Equipment Required

- Leica Digital Level (LS/DNA series)
- Bar-coded staff (preferably Invar staff)
- Tripod
- Open, stable ground with at least **45-60 m length**

10.3 Procedure

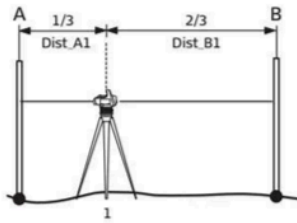
- Ensure staffs are held vertically on firm ground.
- Perform test in calm weather (avoid strong sun, heat shimmer, or wind).
- Always repeat the procedure twice for confirmation.
- Perform collimation checks **before precise levelling projects**

10.3.1 Mark Points

- Fix two staff positions on stable ground about **60 m apart**.
 - Call the first staff position **Point A**.
 - Call the second staff position **Point B**.
 - Mark a point **X** approximately **20 m** from Point A.
 - Mark another point **X'** approximately **20 m** from Point B.
 - So the layout is: **A — X — X' — B**
- Adjustment method A x x B

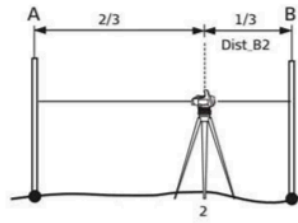
For this adjustment method, the distances to the staffs have a 1:2 ratio for each position of the instrument.

Förstner Procedure



First position (station 1):

- Position the instrument at 1/3 of the distance D between staff A and B.
- The distance D should be approximately 45 m - 60 m.
- The following requirement must be fulfilled: $0.2 \times D < \text{Dist_A1} < 0.4 \times D$ and $D = \text{Dist_A1} + \text{Dist_B1}$.



Second position (station 2):

- Position the instrument at 2/3 of the distance D between staff A and B.
- The following requirement must be fulfilled: $0.2 \times D < \text{Dist_B2} < 0.4 \times D$ and $D = \text{Dist_A1} + \text{Dist_B1}$.

10.3.2 Instrument Setup near A (at X)

1. Set up the instrument at **X** (≈ 20 m from A).
2. Take staff readings at **A** and **B**.
3. Compute the observed height difference (Δh_1).

10.3.3 Instrument Setup near B (at X')

1. Shift the instrument to **X'** (≈ 20 m from B).
2. Take staff readings at **A** and **B**.
3. Compute the observed height difference (Δh_2).

10.3.4 Collimation Error Calculation

- From the midpoint setup, Δh (true difference) is the reference.
- Compare Δh_1 (near A) and Δh_2 (near B) with Δh .
- Any difference indicates **collimation error (e)**.
- Formula:
- $e = 2(\Delta h_1 - \Delta h_2)$

10.3.5 Adjustment

- Enter the **Instrument Menu** → **Check & Adjustment** → **Collimation Test**.
- Follow the guided procedure (Leica will prompt staff positions and readings).
- The instrument computes and stores the correction factor internally.

11 Field Procedure for Levelling line

11.1 Work at First Station

- keep the Levelling Staff at known or (1st) Bench mark
- Levelling instrument will keep centrally between Two Staves.
- In below screen B of BFFB is highlighted in RED color

Line Levelling | ↩

Meas	Code
Backsight:	BFFB BFFB
Stat. ID :	Stn1
PtID :	A1
Rem. :	-----
Height :	0.00000 m

| Rec | Page | Back

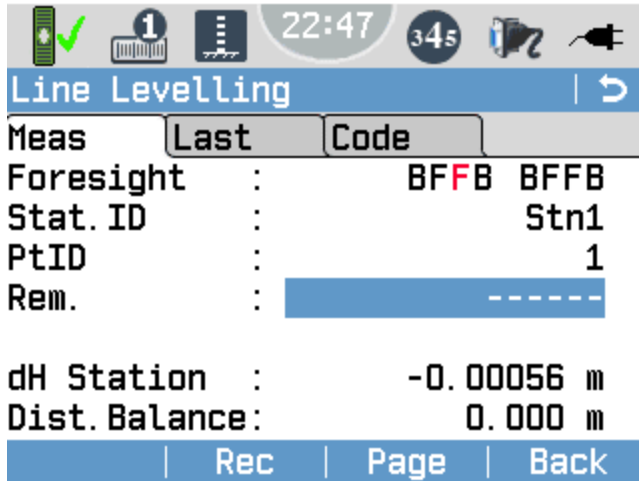
- Here you have to sight backsight point and then press trigger key.

Line Levelling | ↩

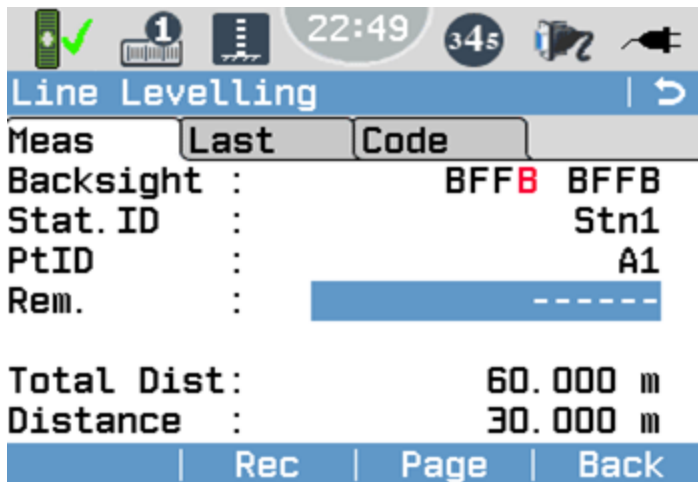
Meas	Last	Code
Foresight :	BFFB BFFB	
Stat. ID :	Stn1	
PtID :	1	
Rem. :	-----	
Total Dist :	30.000 m	
Distance :	30.000 m	
Dist. Balance:	30.000 m	

| Rec | Page | Back

- Here you have to sight foresight point and then press trigger key.
- Now disturb the instrument and set the bubble in centre



- Again, sight foresight point and then press trigger key.



- Now sight back sight point and press trigger
- After completion of First Station, station ID will change into Stn2.
- Now, shift the instrument to the second station

11.2 Work at 2nd and Successive Stations till Last station:

- Now in below screen B of BFFB is highlighted in RED color in second set.
- Here you have to sight backsight point and then press trigger key.

First Backsight Screen (Station2)

The field **Stat.ID** displays the ID of the next station. In the **Backsight/Foresight** field, the first viewing direction of the next station is highlighted in red.

Line Levelling	
Meas	Camera Last Code
Backsight	BFFB BFFB
Stat. ID	Stn2
PtID	POINT01
Rem.	-----
Total dH	0.08264 m
Height	2.58264 m
Dist. Balance	0.000 m
Focus	Rec Page ↓
SetOut	INT Station ↓
Back	Check Close ↑

Total dH: Height difference between start and current backsight.

Height: Height of current backsight.

- Then you have to sight foresight point and then press trigger key.

Now disturb the instrument and set the bubble in centre

- Again, sight foresight point and then press trigger key.
- Now sight back sight point and press trigger
- After completion of First Station, station ID will change into Stn3
- Similarly, Continue levelling process and close the levelling line at known BM (Bench Mark).
- To check/see Total Distance & Total H. change the page by **Arrow** Key two times
- And then Select **Close**.

15:21 34s

Line Levelling | ↩

Meas	Last	Code
Backsight :	BFFB	BFFB
Stat. ID :		Stn2
PtID :		1
Rem. :	-----	
Total dH :	-0.0004	m
Height :	-0.0004	m
Dist. Balance:	0.000	m

Back | Check | Close | ↑

15:24 34s

Line Info | ↩

Info

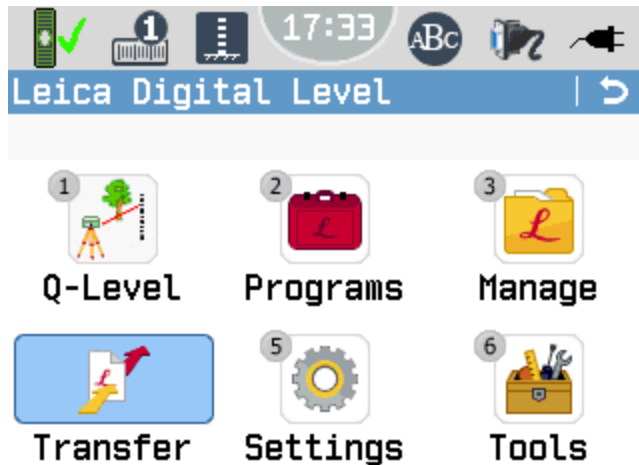
Line ID :	LINE00001
No. Station :	1
Total dH :	-0.0004 m
Total Dist :	60.000 m
Dist Balance:	0.000 m

Back | | | Cont

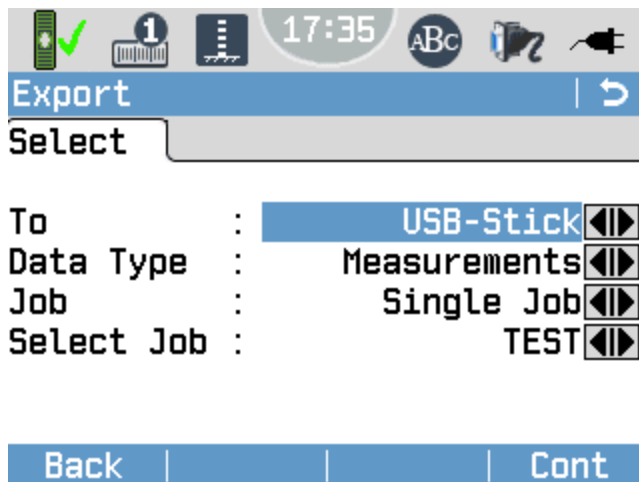
- And check your desired values Total Distance and height then click on **Back(F1)** Button.

12 Data Export

- Select **Transfer** in the main menu (**HOME**) and press **ok**.



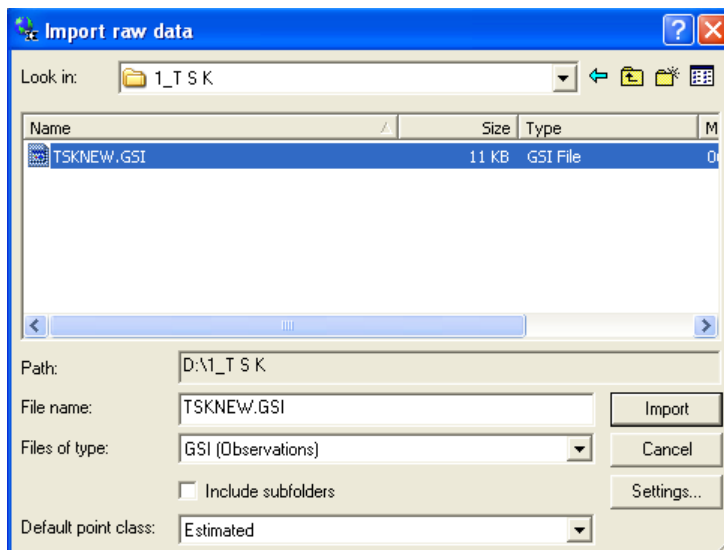
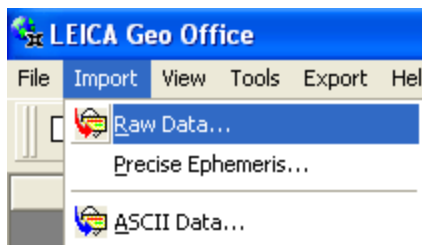
- Select **Export**
- Set as below option and **Cont.**

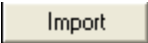


- Press **F1** the Select required Format like ASCII or GSI format. Then Continue (**F4**).
- Data will be successfully Exported to **USB- Stick**.

13 Processing Digital Levelling Data Using Leica Geo Office

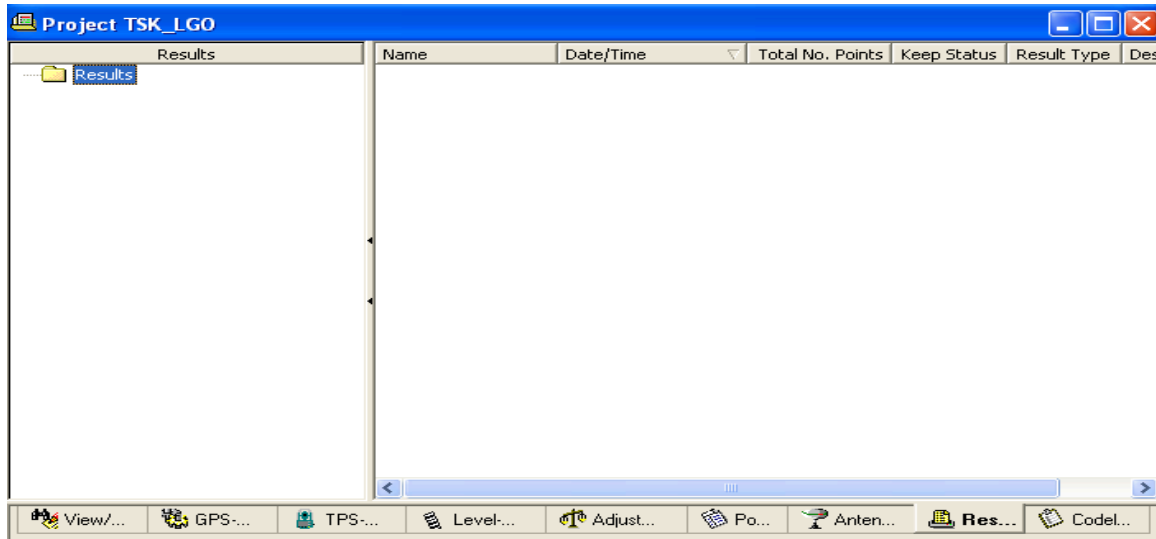
- Start Leica Geo Office (LGO) v5.0 with the hardware lock (dongle) inserted. The **license number**, **maintenance number**, and the **hardware lock number** are each unique and tied to the software CD. **Do not interchange hardware locks** between different licenses or CDs.
- Import the *.GSI data into LGO v5.0 for processing.(Ensure that the observed *.GSI data has been stored on the desktop PC beforehand.)

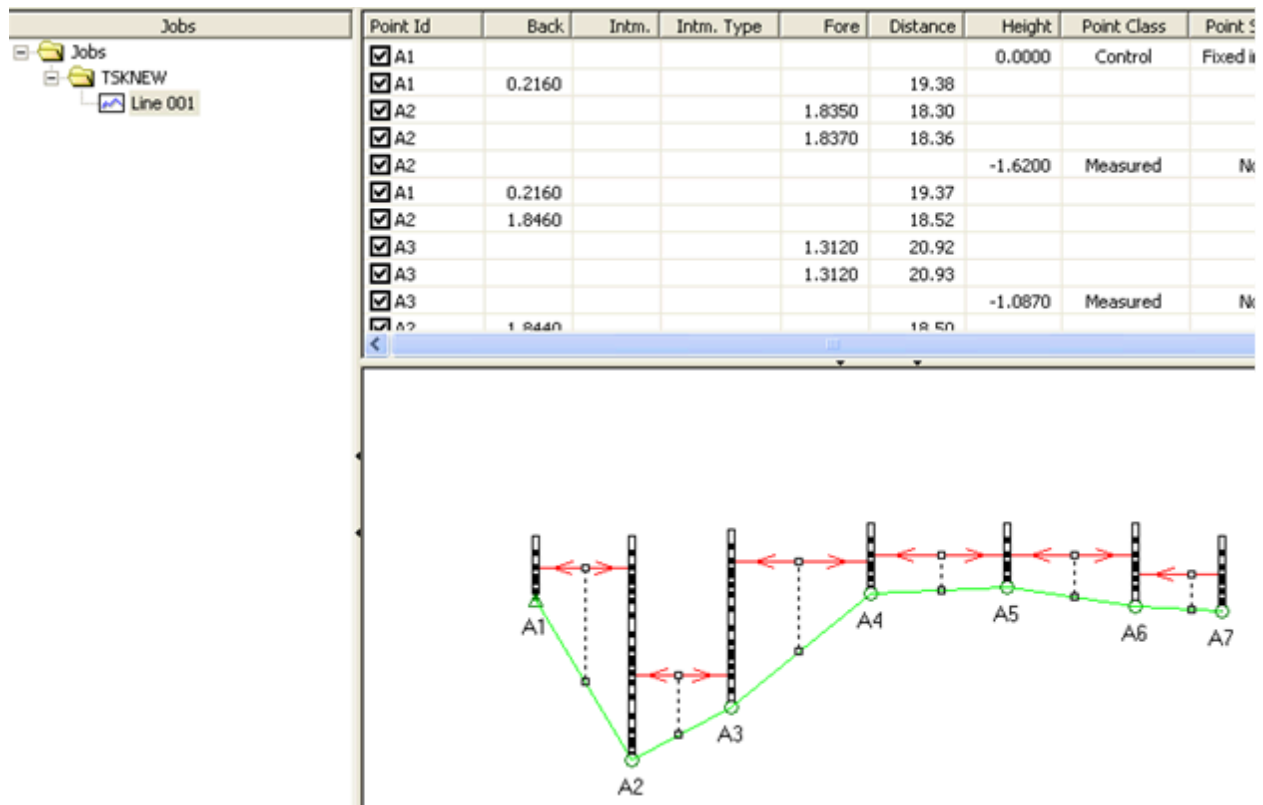


- Select the path and file where *.GSI is stored.
- File of Type: GSI (Observations).
- Default point class: Estimated
- Click 
- A progress bar appears for few seconds.

- “Assign data to Project” opens
- Place the cursor on the project folder,
- Right click New enter Project Name click OK and click  and Close.

Dialog with Project name opens (TSK_LGO is project name assigned)





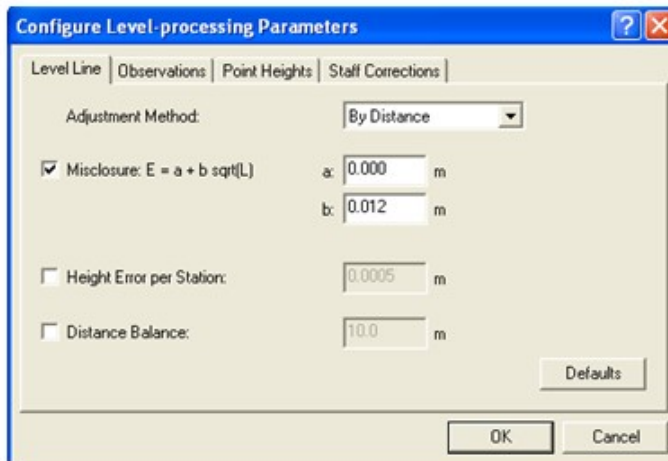
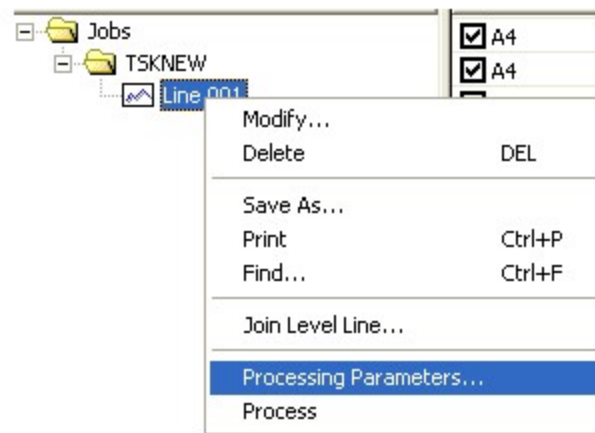
Open the **Jobs Tree** and select the **Level Line**.

Select the **first point**, right-click on the **Height** column, and modify the height from **0.000** to the control height (e.g., **498.434**).

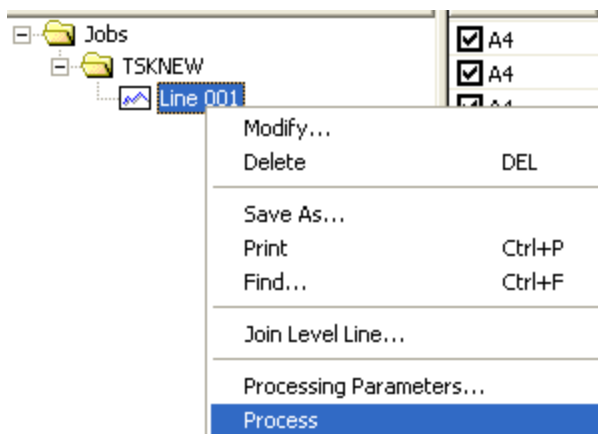
Similarly, select the **last observed point**, right-click on the **Height** column, and modify the height from **0.000** to the control height (e.g., **498.317**).

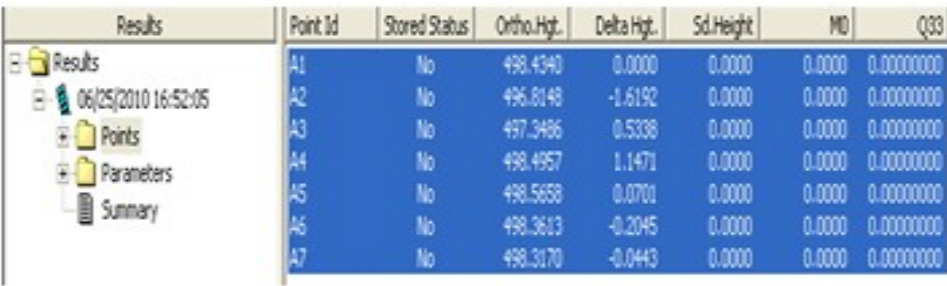
Next, select the **last observed point**, right-click on **Point Class**, and choose **Create Control**. The point class will now change to **Fixed in Height**.

Right click on the level line and select Processing Parameters:



Select Adjustment method as "By Distance".
 Misclosure: $a = 0.000$
 $B = 0.012$ (as required)
 Check off Height error per station.
 Check off Distance Balance.
 Click **OK**
 Dialog box disappears.





above dialog opens.

DoubleClick on  Summary to view the results of leveling line.

Point Id	Stored Status	Ortho.Hgt.	Delta Hgt.	Sd.Height	MD	Q33
A1	No	498.4340	0.0000	0.0000	0.0000	0.00000000
A2	No	496.8148	-1.6192	0.0000	0.0000	0.00000000
A3	No	497.3486	0.5338	0.0000	0.0000	0.00000000
A4	No	498.4957	1.1471	0.0000	0.0000	0.00000000
A5	No	498.5658	0.0701	0.0000	0.0000	0.00000000
A6	No	498.3613	-0.2045	0.0000	0.0000	0.00000000
A7	No	498.3170	-0.0443	0.0000	0.0000	0.00000000

13.1.1 Generate Report

1. Go to **Reports > Leveling Report**.
2. Choose the format (adjusted heights, residuals, misclosure).
3. Export the report to PDF, Word, or Excel.

Results will be displayed as below



Level Summary

TSK_LGO

06/25/2010 16:54:38

Project Information

Project name: TSK_LGO
Date created: 06/25/2010 16:24:51
Application software: LEICA Geo Office 5.0

Line 001

Line length: 270.6030 m
Method: BFFB
Total station difference: 0.0070 m
Start point id: A1
Number of stations: 6
Date/time: 06/25/2010 10:32:50
Number of observations: 24

Processing Parameters

Adjustment method: by distance
Processed with Staff
Corrections: No
Height difference: -0.1225 m

Tolerance	Permitted [m]	Actual [m]	Accepted
Misclosure	0.0062	-0.0055	✓

Points

Point Id	Epoch	Height [m]	Corr [m]	Delta Hgt. [m]	Point Class	Sd. Hgt. [m]
A1	06/25/2010 10:32:51	498.4340	-	-	Control	-
A2	06/25/2010 10:32:56	496.8148	498.4348	-1.6192	Measured	-
A3	06/25/2010 10:33:02	497.3486	498.4356	0.5338	Measured	-
A4	06/25/2010 10:33:08	498.4957	498.4367	1.1471	Measured	-
A5	06/25/2010 10:33:14	498.5658	498.4378	0.0701	Measured	-
A6	06/25/2010 10:33:20	498.3613	498.4383	-0.2045	Measured	-
A7	06/25/2010 10:33:26	498.3170	-	-0.0443	Control	-

Height(m): Is the corrected height in meter; Corr(m):
(Height Residual = H_c+correction;)

Delta ht: Corrected level reduced from the last station;

Height (m)=H_o+ Sum of LastDelta Hgt.

(Example: A3 corrected height = 498.434+(-

1.6192+0.5338)=497.3486;

A6=498.434+(-1.6192+0.5338+1.1471+0.0701-
0.2045)=498.3613)

The data can be selected with mouse right click and print.

13.1.2 Save and Archive Project

- Save the project in LGO.
- Backup the data and reports for documentation.